SCM 518 – Analytical Decision Modeling

Final Project Report

Title: Optimizing E-Scoot Charging Infrastructure at ASU Tempe Campus

Team No: M6

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I. INTRODUCTION

Electric scooters (E-Scoot) have become a popular mode of transport at ASU, offering a fast and eco-friendly way to navigate the large Tempe campus. However, the growing number of E-Scoot users has outpaced the availability of charging stations, leading to inconvenience and limiting the university's sustainability goals.

This project aims to help ASU make data-driven decisions on where to install E-Scoot charging stations and how many to place at each location. Using optimization techniques from SCM 518 and tools like Python and Gurobi, we developed a model that considers cost, demand, and accessibility to recommend the best deployment strategy.

Our model ensures that all key campus zones are covered, the budget is respected, and the infrastructure supports broader adoption of sustainable commuting.

II. PROJECT OBJECTIVE

The goal of this project is to develop a practical, cost-effective plan for deploying E-Scoot charging stations across ASU's Tempe campus using an optimization-based approach. The model is designed to help campus planners answer key questions such as:

1. **Which locations should be selected?** Identify high-need areas that are accessible and capable of serving multiple demand zones—including dorms, classrooms, and recreation spaces.
2. **How many chargers should be installed per location?** Allocate the right number of chargers based on local demand—ensuring each site is properly equipped without overspending.
3. **Can we stay within budget?** Ensure the total cost of installations (fixed and variable) remains under the $35,000 limit.
4. **Is the solution fair?** Guarantee that at least 70% of residential charging demand is met by nearby stations.
5. **Is usage balanced?** Limit any single station from serving more than 30 users to avoid overloading.

Beyond optimization, this project highlights how analytical models can enhance decision-making, promote sustainability, and guide strategic planning for smart campus infrastructure.

III. BACKGROUND & PROBLEM OVERVIEW

With E-Scoot usage rapidly growing at ASU, there is increasing pressure to expand the campus's charging infrastructure. The Tempe campus covers a large area, and students regularly travel between dorms, classrooms, gyms, and other facilities. Yet, reliable charging access remains limited, creating a bottleneck for sustainable commuting.

Installing charging stations is not a simple task. Each location has associated fixed and variable costs, and planners must make decisions within a $35,000 budget. High-traffic areas may require more chargers, while some residential zones need guaranteed coverage—even if they're less central. Striking the right balance between cost, accessibility, and fairness is essential.

This project addresses the challenge as a classic **facility location problem**, using optimization techniques to determine:

- Where chargers should be installed
- How many units each location should host
- How to cover all demand zones without exceeding budget

Beyond minimizing cost, the model incorporates practical planning constraints: ensuring residential zones are prioritized, preventing any site from being overloaded, and fairly distributing infrastructure across campus. The result is a data-driven tool that helps ASU plan a smarter, greener future using real-world analytical methods.

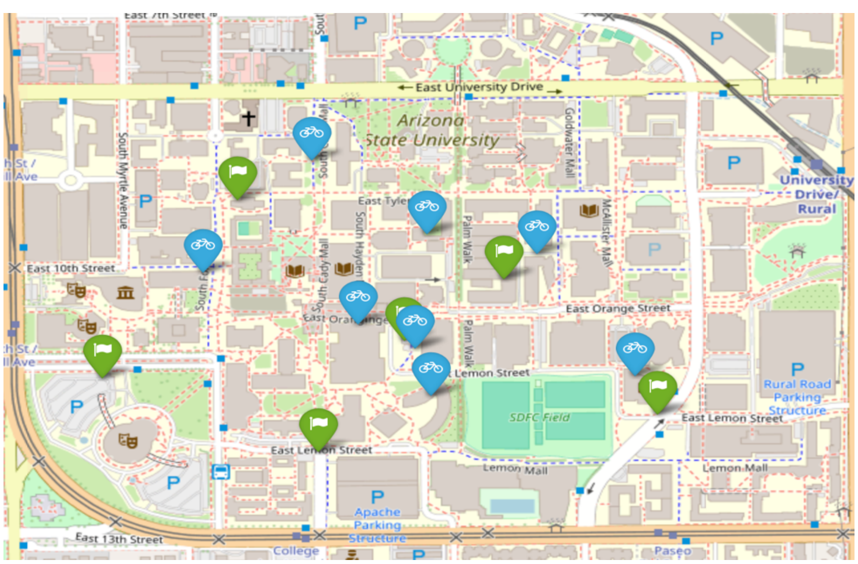
IV. METHODOLOGY & TOOLS

We followed a structured, data-driven approach to develop an optimization model for E-Scoot charger placement at ASU:

1. **Identifying Locations & Zones**
   Eight candidate locations were selected based on campus layout, including libraries, dorms, and academic centers. Six high-demand zones were defined based on student density and movement patterns.
2. **Data Estimation**
   Demand was estimated using logical assumptions. Costs were divided into fixed (per station) and variable (per charger), using online references. Coverage was determined using walking distances from Google Maps.
3. **Model Formulation**
   We used binary variables for station placement and integer variables for charger counts. Constraints addressed demand fulfillment, budget limits, dormitory fairness (70% coverage), and max capacity (≤30 users per station). The goal was to minimize total installation cost.
4. **Tools Used**
   - Python & Gurobi for modeling and solving
   - Excel for organizing input data
   - Google Maps to define location-to-zone coverage
5. **Testing & Validation**
   The model was tested under different scenarios to ensure reliability, and outputs were mapped to verify campus-wide coverage.

This approach ensured the model was both practical and aligned with real planning needs at ASU.

V. LOCATIONS & ZONES (DETAILED DATA)



We considered 8 possible station locations and 6 demand zones.

Charging Station Locations:

Location	Fixed Cost ($)	Variable Cost/Charger ($)	Max Chargers	Type
Hayden Library	3200	600	6	Academic
Hassayampa Dormitory	3500	650	9	Residential
Sun Devil Fitness Center	2800	700	7	Recreation
MU Food Court	3000	750	5	Food Court
Palo Verde Dormitory	2900	720	8	Residential
Engineering Center	3600	580	7	Academic
Health Services	3400	630	8	Service
W P Carey Business School	3100	670	5	Academic

Demand Zones and Estimated Demand:

Zone Name	Demand (Number of Students Needing Charge)
North Campus Housing	9
South Campus Housing	6
Academic Core	13
Athletic Facilities	11
Memorial Union/Central	14
Business District	7

Coverage Matrix (Which Locations Serve Which Zones):

Location	North Campus Housing	South Campus Housing	Academic Core	Athletic Facilities	MU/Central	Business District
Hayden Library	0	1	1	0	1	1
Hassayampa Dormitory	1	1	0	0	1	0
Sun Devil Fitness Center	0	0	1	1	1	0
MU Food Court	1	0	1	0	1	0
Palo Verde Dormitory	1	1	0	0	1	0
Engineering Center	0	1	1	1	0	1
Health Services	1	0	1	1	1	1
W P Carey Business School	0	0	1	0	1	1

VI. OPTIMIZATION MODEL

Inputs:

$i \in L$: index for Locations (ex. L_1 = Hayden Library, L_2 = Hassayampa Dormitory, L_3 = Sun Devil Fitness Centre, L_4 = MU Food Court, L_5 = Palo Verde Dormitory, L_6 =Engineering Center, L_7 = Health Services, L_8 = W P Carey Business School)

$j \in Z$: index for Demand Zones (ex. Z_1 = North Campus Zone 1, Z_2 = South Campus Zone, Z_3 = Academic Core, Z_4 = Athletic Facilities, Z_5 = MU/Central, Z_6 =Business District)

f_i = Fixed Cost for installing station at location i
v_i = Variable cost per charging point at location i
M_i = Maximum no. of charging points allowed at location i
d_j = Charging demand (no. of students) in zone j
$a_{ij} \in \{0,1\}$: If location i covers demand zone j then 1 else 0

Decision Variables:

$y_i \in \{0,1\}$: if a station is installed at location i then 1 else 0
x_i = No of charging points installed at location i

Objective:

Minimize Z = $\sum_{i \in L} (f_i * y_i + v_i * x_i)$

Constraints:

1. Each zone's demand must be fully served:

$$\sum_{i \in L} a_{ij} * x_i \geq d_j$$

2. Can't install chargers unless station exists:

$$x_i \leq M_i * y_i$$

3. Budget constraint:

$$\sum_{i \in L} (f_i * y_i + v_i * x_i) \leq 35000$$

4. Residential Coverage Constraint: At least 70% of North Campus Zone and South Campus Zone demand must be served by residential locations.

$$x_{L2} + x_{L5} \geq 0.7(d_{Z1} + d_{Z2})$$

5. Load Balancing Constraint: No single location should serve more than 50% of total demand (30 students)

$$\sum_{j \in Z} (a_{ij} * x_i) \leq 30$$

6. Non-negative constraint:

$$x_i \geq 0$$

7. Binary Constraint

$$y_i \in \{0,1\}$$

VII. MODEL OUTPUT & INTERPRETATION

After solving our optimization model using Gurobi in Python, we arrived at a data-backed solution that identifies the most cost-effective and strategic way to place E-Scoot chargers across the ASU Tempe campus.

The output not only meets budget constraints but also ensures fair coverage across high-demand residential and academic areas. Below is a tabular summary of the final plan:

```
Charging Station Deployment Plan (ASU Tempe Campus):
------------------------------------------------------------
Location                           Installed    Chargers
------------------------------------------------------------
Hayden Library                     0            0
Hassayampa Dormitory               1            9
Sun Devil Fitness Center           1            6
MU Food Court                      0            0
Palo Verde Dormitory               1            2
Engineering Center                 1            7
Health Services                    0            0
W. P. Carey Business School        0            0
------------------------------------------------------------

Total Installation Cost = $28,350.00
```

Key Observations:

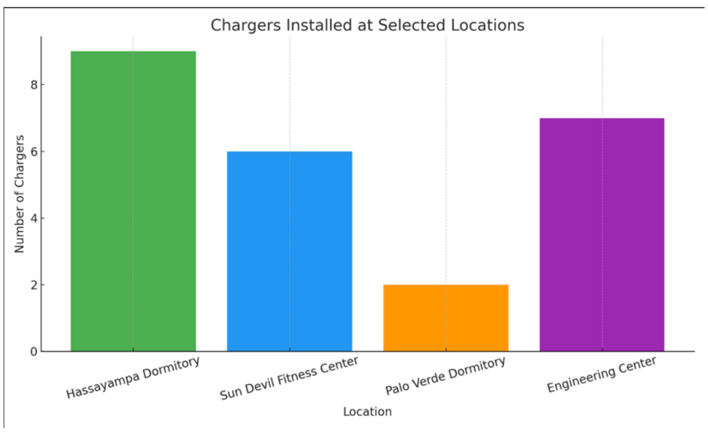
• **Budget Efficiency:** The plan used only $28,350, staying $6,650 under budget, allowing room for potential future expansion.

• **Residential Fairness Constraint Met:** Hassayampa and Palo Verde Dormitories alone meet the requirement of covering at least 70% of housing demand.

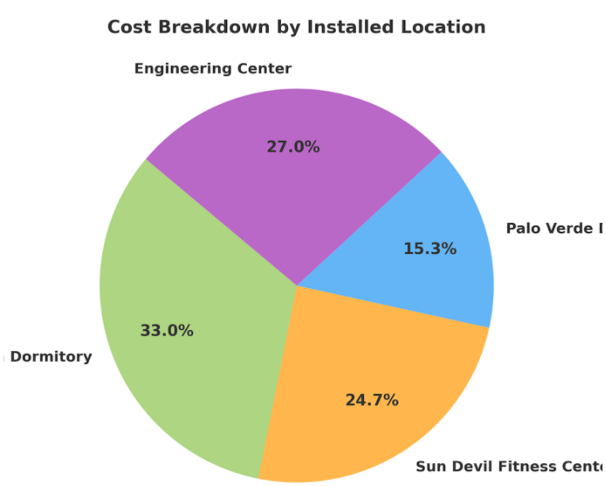
• **Strategic Site Selection:** Only 4 out of 8 locations were activated, avoiding unnecessary installations and maximizing coverage.

• **No Overload:** All locations adhere to the constraint of serving no more than 30 users.

• **Full Coverage:** All 6 zones received enough chargers to meet or exceed their demand levels.



This bar chart shows that Hassayampa Dormitory received the most chargers (9), followed by Engineering Center (7), Sun Devil Fitness Center (6), and Palo Verde Dormitory (2). These sites were chosen based on their strategic ability to cover surrounding high-demand zones.



This pie chart highlights that Hassayampa Dormitory consumed the largest share (33%) of the total installation budget, with Engineering Center (27%), Sun Devil Fitness Center (25%), and Palo Verde Dormitory (15%) following. This distribution aligns with the number of chargers installed and their respective unit costs.

VIII. ANSWER REPORT

Microsoft Excel 16.96 Answer Report
Worksheet: [M6 Excel E-scoot Charging Optimization.xlsx]General Info
Report Created: 5/3/25 11:46:51
Result: Solver found an integer solution within tolerance. All constraints are satisfied.
Solver Engine
Solver Options

Objective Cell (Min)

Cell	Name	Original Value	Final Value
$B$28	Total Installation Cost ($) L1	28350	28350

Variable Cells

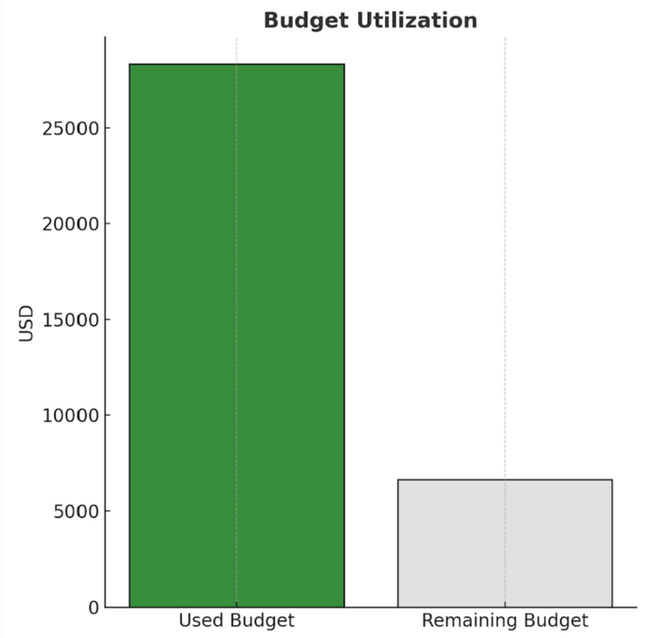
Cell	Name	Original Value	Final Value	Integer
$F$5:$G$12				

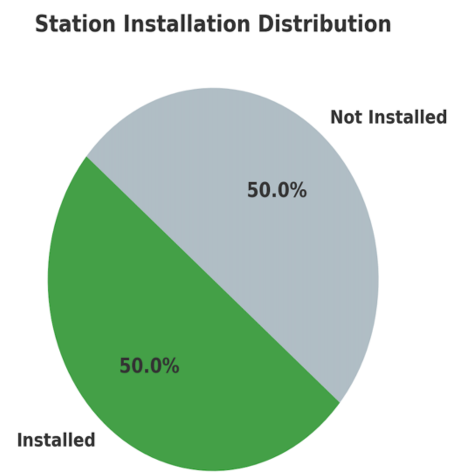
Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$B$28	Total Installation Cost ($) L1	28350	$B$28<=$B$1	Not Binding	6650
$D$27	L2+L5 L3	11	$D$27>=$D$29	Not Binding	0.5
$G$5:$G$12 <= $H$5:$H$12					
$K$16:$K$21 <= $M$16:$M$21					
$K$16:$K$21 >= $M$5:$M$10					
$F$5:$F$12=Binary					
$G$5:$G$12=Integer					

- **Solver Objective Value**: Total cost = $28,350
- **Budget status**: Constraint is **not binding** → ample unused budget ($6,650 left)
- **Fairness status**: Minimum 70% dorm coverage rule is **met** but **close to threshold** (only 0.5 above requirement)
- **Feasibility**: Solver confirms **all constraints are satisfied**
- **No constraint is binding** → there's still room to adjust the solution if needed

We used two visuals to help interpret these effects:





- **Budget Utilization**: The bar chart shows we used $28,350 of the $35,000 budget, leaving a comfortable margin. This proves the model is cost-conscious and has room for future flexibility.
- **Installation Distribution**: The pie chart reveals an even split—only 4 out of 8 candidate locations were selected. This means the model was strategic, not wasteful, when choosing charger sites.

IX. CHALLENGES AND LIMITATIONS

- Coverage data had to be manually built from map estimates
- Real installation costs were approximated based on public sources
- Demand forecasts lacked historical usage data, so estimates were based on foot traffic patterns
- Solver performance varied with changes in coverage density and budget

X. KEY LEARNINGS & TAKEAWAYS

This project helped us understand how optimization can improve real-world decisions. Key takeaways include:

- **Modeling makes planning easier**: Turning a complex problem into a structured model helped us find the best solution quickly and logically.
- **Constraints are valuable**: Budget limits, demand needs, and fairness rules guided our decisions and made the outcome more practical.

- **Tools like Python and Gurobi are powerful**: They made solving a complex setup faster and more manageable.
- **Adaptability matters**: The model responded well to changing inputs, showing how flexible planning is key to real-world success.
- **Smart planning balances cost and fairness**: We optimized expenses without ignoring dorm needs, proving sustainability and equity can go together.

XI. TEAM CONTRIBUTION

Team Member	Contribution (%)
Vinit Vijaykumar Adke	25%
Yash Sanjaykumar Pardeshi	25%
Jaswanth Giridhargopalan	25%
Dhirraj Suresh Kumar	25%

XII. APPENDICES

- Appendix A: Python code used for model implementation

Step 1: Input Data

We define the problem inputs including potential locations, demand zones, costs, capacity, demand, and the coverage matrix.

```
# Import necessary library
from gurobipy import Model, GRB, quicksum

# Define candidate locations and zones
locations = ['L1', 'L2', 'L3', 'L4', 'L5', 'L6', 'L7', 'L8']
zones = ['Z1', 'Z2', 'Z3', 'Z4', 'Z5', 'Z6']

# Define costs
fixed_cost = {'L1': 3200, 'L2': 3500, 'L3': 2800, 'L4': 3000,
              'L5': 2900, 'L6': 3600, 'L7': 3400, 'L8': 3100}
variable_cost = {'L1': 600, 'L2': 650, 'L3': 700, 'L4': 750,
                 'L5': 720, 'L6': 580, 'L7': 630, 'L8': 670}
max_chargers = {'L1': 6, 'L2': 9, 'L3': 7, 'L4': 5,
                'L5': 8, 'L6': 7, 'L7': 8, 'L8': 5}

# Demand in each zone
demand = {'Z1': 9, 'Z2': 6, 'Z3': 13, 'Z4': 11, 'Z5': 14, 'Z6': 7}
total_demand = sum(demand.values())

# Coverage matrix
coverage = {
    ('L1','Z1'):0, ('L1','Z2'):1, ('L1','Z3'):1, ('L1','Z4'):0, ('L1','Z5'):1, ('L1','Z6'):1,
    ('L2','Z1'):1, ('L2','Z2'):1, ('L2','Z3'):0, ('L2','Z4'):0, ('L2','Z5'):1, ('L2','Z6'):0,
    ('L3','Z1'):0, ('L3','Z2'):0, ('L3','Z3'):1, ('L3','Z4'):1, ('L3','Z5'):1, ('L3','Z6'):0,
    ('L4','Z1'):1, ('L4','Z2'):0, ('L4','Z3'):1, ('L4','Z4'):0, ('L4','Z5'):1, ('L4','Z6'):0,
    ('L5','Z1'):1, ('L5','Z2'):1, ('L5','Z3'):0, ('L5','Z4'):0, ('L5','Z5'):1, ('L5','Z6'):0,
    ('L6','Z1'):0, ('L6','Z2'):1, ('L6','Z3'):1, ('L6','Z4'):1, ('L6','Z5'):0, ('L6','Z6'):1,
    ('L7','Z1'):1, ('L7','Z2'):0, ('L7','Z3'):1, ('L7','Z4'):1, ('L7','Z5'):1, ('L7','Z6'):1,
    ('L8','Z1'):0, ('L8','Z2'):0, ('L8','Z3'):1, ('L8','Z4'):0, ('L8','Z5'):1, ('L8','Z6'):1,
}

# Budget limit
budget = 35000
```

Step 2: Define Decision Variables

We create integer and binary variables for the number of chargers and installation status respectively.

```
# Create the model
model = Model("Ebike_Charging_Station_Optimization")

# Integer variable for number of chargers
x = model.addVars(locations, vtype=GRB.INTEGER, name="ChargersInstalled")

# Binary variable for whether a station is installed
y = model.addVars(locations, vtype=GRB.BINARY, name="StationOpen")
```

Step 3: Define Objective Function

We minimize the total installation cost, which includes both fixed and variable components.

```
# Objective: Minimize total installation cost
model.setObjective(quicksum(fixed_cost[i]*y[i] + variable_cost[i]*x[i] for i in locations), GRB
```

Step 4: Add Constraints

We add five sets of constraints: installation logic, zone demand, budget, dorm fairness, and load balancing.

```
# Constraint 1: Chargers can only be installed if the station is installed
for i in locations:
    model.addConstr(x[i] <= max_chargers[i] * y[i], name=f"MaxCap_{i}")

# Constraint 2: Each zone must meet or exceed its demand
for j in zones:
    model.addConstr(quicksum(coverage[i, j] * x[i] for i in locations) >= demand[j], name=f"Dem

# Constraint 3: Total installation cost must not exceed budget
model.addConstr(quicksum(fixed_cost[i] * y[i] + variable_cost[i] * x[i] for i in locations) <=

# Constraint 4: At least 70% of dorm area demand must be covered by L2 and L5
model.addConstr(x['L2'] + x['L5'] >= 0.7 * (demand['Z1'] + demand['Z2']), name="ResidentialCove

# Constraint 5: No single location serves more than 30 students
for i in locations:
    model.addConstr(quicksum(coverage[i, j] * x[i] for j in zones) <= 30, name=f"LoadBalance_{i
```

Step 5: Solve and Display Results

We solve the model and then print the decision for each location in a readable format.

```
# Optimize the model
model.optimize()

# Location names
asu_locations = {
    'L1': "Hayden Library", 'L2': "Hassayampa Dormitory", 'L3': "Sun Devil Fitness Center",
    'L4': "MU Food Court", 'L5': "Palo Verde Dormitory", 'L6': "Engineering Center",
    'L7': "Health Services", 'L8': "W. P. Carey Business School"
}

# Print output
print("\n📊 Charging Station Deployment Plan (ASU Tempe Campus):")
print("-" * 65)
print(f"{'Location':<35} {'Installed':<12} {'Chargers':<10}")
print("-" * 65)
for i in locations:
    print(f"{asu_locations[i]:<35} {int(y[i].X):<12} {int(x[i].X):<10}")
print("-" * 65)
print(f"\n✅ Total Installation Cost = ${model.ObjVal:,.2f}")
```

```
Charging Station Deployment Plan (ASU Tempe Campus):
-----------------------------------------------------------------
Location                            Installed    Chargers
-----------------------------------------------------------------
Hayden Library                      0            0
Hassayampa Dormitory                1            9
Sun Devil Fitness Center            1            6
MU Food Court                       0            0
Palo Verde Dormitory                1            2
Engineering Center                  1            7
Health Services                     0            0
W. P. Carey Business School         0            0
-----------------------------------------------------------------

Total Installation Cost = $28,350.00
```

- Appendix B: Excel & Solver Screenshots

Total Budget ($)	35000

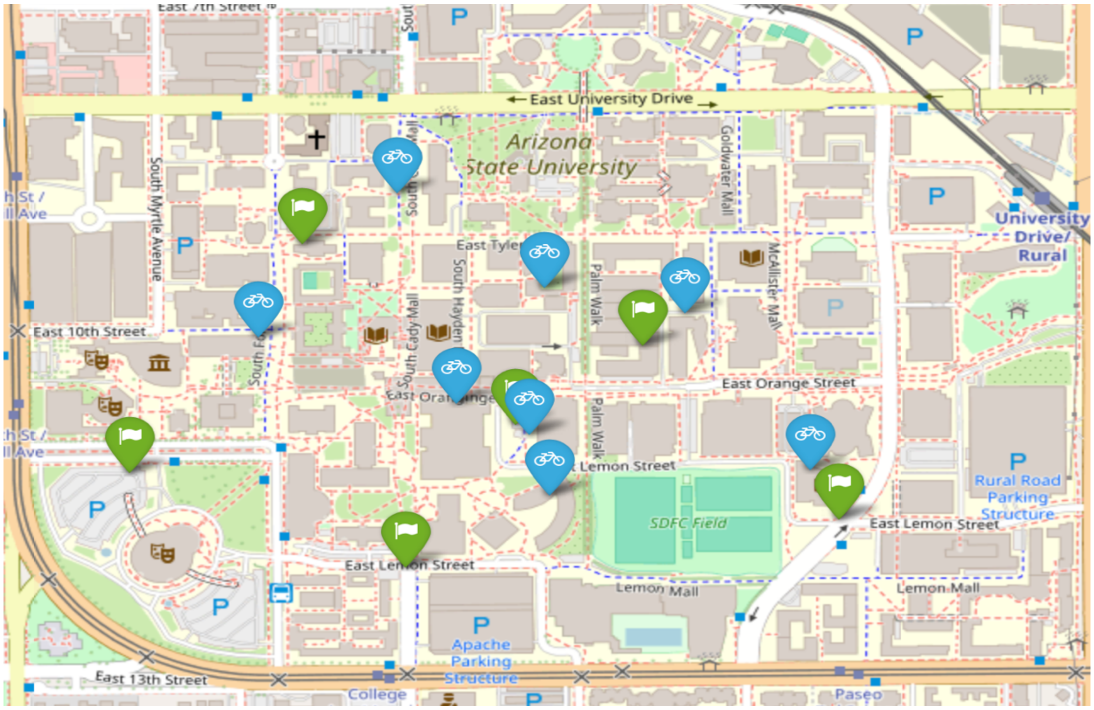
	i	fi	vi	Mi	yi	xi	Constraint
Location Code	Location Name	Fixed Cost ($)	Variable Cost/Charger ($)	Max Chargers	Station Open (Binary)	No of chargers installed	Mi*xi
L1	Hayden Library	3200	600	6	0	0	0
L2	Hassayampa Dormitory	3500	650	9	1	9	9
L3	Sun Devil Fitness Center	2800	700	7	1	6	7
L4	MU Food Court	3000	750	5	0	0	0
L5	Palo Verde Dormitory	2900	720	8	1	2	8
L6	Engineering Center	3600	580	7	1	7	7
L7	Health Services	3400	630	8	0	0	0
L8	W P Carey Business School	3100	670	5	0	0	0

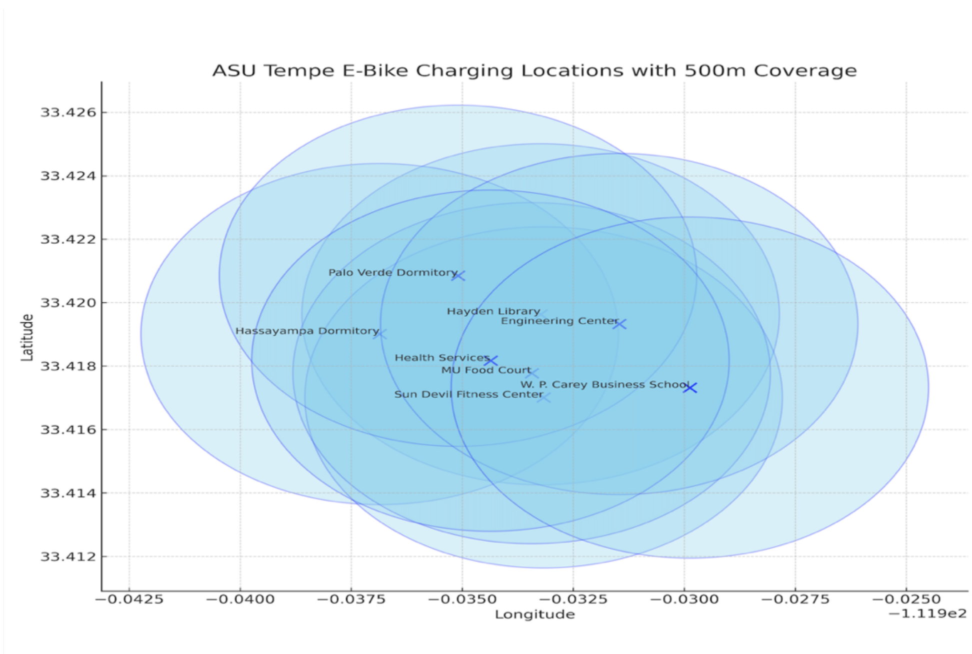
	j	dj
Zone Code	Zone Name	Demand
Z1	North Campus Housing	9
Z2	South Campus Housing	6
Z3	Academic Core	13
Z4	Athletic Facilities	11
Z5	Memorial Union/Central	14
Z6	Business District	7

Location Code	L1	L2	L3	L4	L5	L6	L7	L8	aij*xi		
North Campus Housing (Z1)	0	1	0	1	1	0	1	0	11	<=	30
South Campus Housing (Z2)	1	1	0	0	1	1	0	0	18	<=	30
Academic Core (Z3)	1	0	1	1	0	1	1	1	13	<=	30
Athletic Facilites (Z4)	0	0	1	0	0	1	1	0	13	<=	30
Memorial Union/Central (Z5)	1	1	1	1	1	0	1	1	17	<=	30
Business District (Z6)	1	0	0	0	0	1	1	1	7	<=	30

		L2+L5	11
Total Installation Cost ($)	28350		>=
		0.7*(Z1+Z2)	10.5

- Appendix C: Map Visualization





- Appendix D: Visualizations: